X(3872) Analysis Update

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2026.8.27









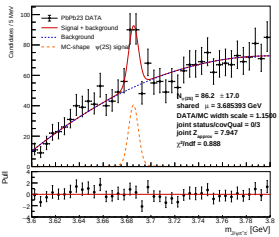

Three comments addressed

Comment	Current status
Add PbPb23 to the $\psi(2S)$ validation	Two-year fit, sPlot and combined closure completed; the validation remains statistics-limited
Constrain the Gaussian widths with MC	Year-specific reweighted-MC shapes define the current nominal DATA fit; fit boundaries remain diagnostics
Compare PbPb23 and PbPb24 signal MC	Pre-ML, unit-normalized year comparisons completed for both X and $\psi(2S)$

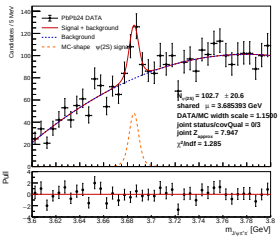
- ppRef vs PbPb MC comparisons are part of the nominal reweighting workflow.
- The two channels show different PbPb23–PbPb24 MC agreement.

PbPb23 added to the $\psi(2S)$ validation

PbPb23 DATA



PbPb24 DATA



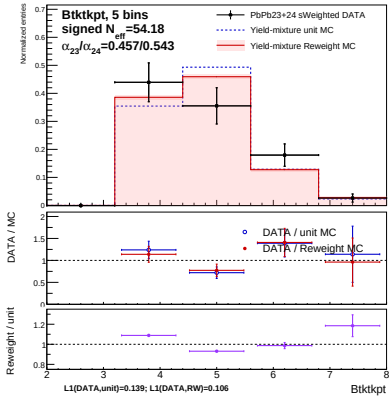
	PbPb23	PbPb24	combined
fitted signal yield	86.32	102.69	189.01
signed N_{eff}	27.759	26.970	54.176

$N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$ for the signed signal sWeights. The 30% point maximizes the combined N_{eff} in the closure scan.

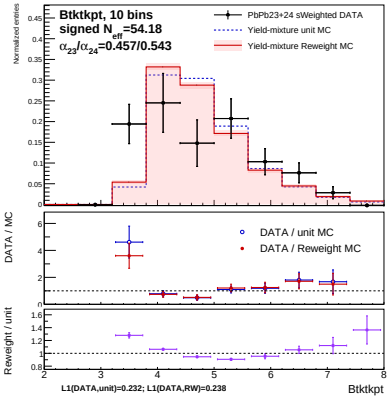
Adding 2023 approximately doubles the combined effective sample size.

$\psi(2S)$ validation: Btktkpt

5 bins



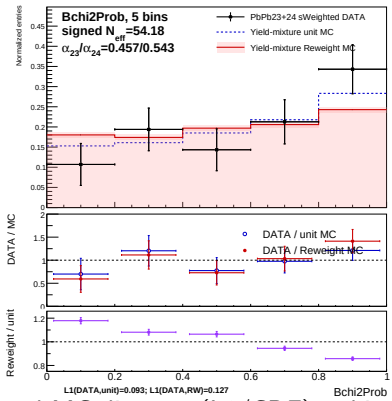
10 bins



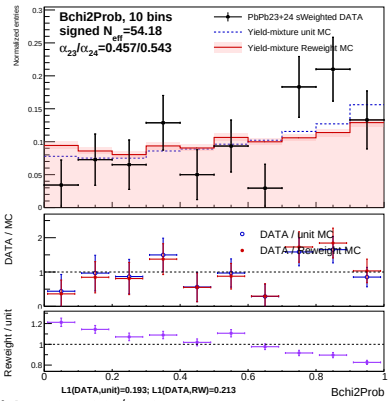
Reweighted-MC distance (L1/CDF): 5 bins 0.106/0.053; 10 bins 0.238/0.140.

$\psi(2S)$ validation: Bchi2Prob

5 bins



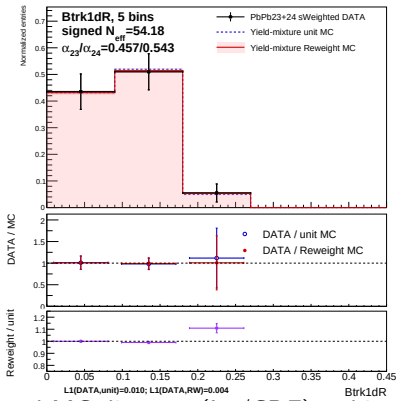
10 bins



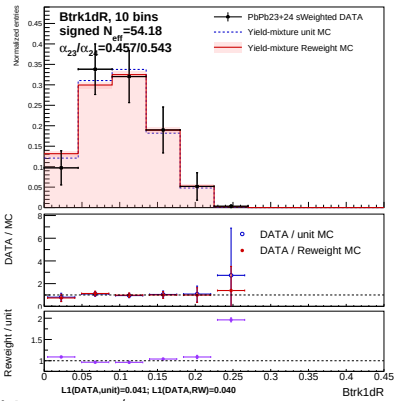
Reweighted-MC distance (L1/CDF): 5 bins 0.127/0.107; 10 bins 0.213/0.177.

$\psi(2S)$ validation: Btrk1dR

5 bins



10 bins



Reweighted-MC distance (L1/CDF): 5 bins 0.004/0.004; 10 bins 0.040/0.035.

X baseline: reweighting and ML recipe

Reweighting

Variables:

- Bcos_dtheta, Btktkpt, Bchi2Prob
- Btrk1Pt, Btrk2Pt, Btrk1dR

Selection:

- $10 < \texttt{Bpt} < 50$, $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$
- $0.9 < \texttt{Btrk1Pt}, \texttt{Btrk2Pt} \leq 4.5$
- $2 \leq \texttt{Btktkpt} \leq 8$, $0 \leq \texttt{Bchi2Prob} \leq 1$
- $0 \leq \texttt{Btrk1dR} \leq 0.45$, $0 \leq \texttt{Btrk2dR} \leq 0.40$
- $-1 \leq \texttt{Bcos_dtheta} \leq 1$

ML

Variables:

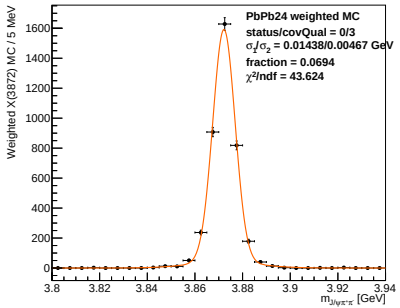
- Btrk1dR, Btktkpt, Btrk1Pt
- Bchi2Prob, Bcos_dtheta, Btrk2Pt

Selection:

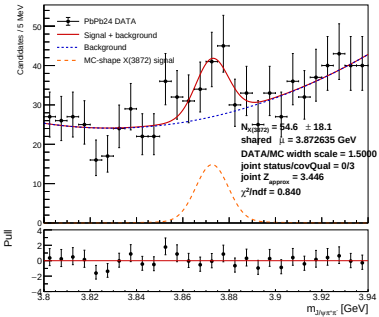
- $10 < \texttt{Bpt} < 50$, $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$
- $0.9 < \texttt{Btrk1Pt}, \texttt{Btrk2Pt} \leq 4.5$
- $2 \leq \texttt{Btktkpt} \leq 8$, $0 \leq \texttt{Bchi2Prob} \leq 1$
- $0 \leq \texttt{Btrk1dR} \leq 0.45$, $0 \leq \texttt{Btrk2dR} < 0.25$
- $-1 \leq \texttt{Bcos_dtheta} \leq 1$
- DATA background: $3.82 < \texttt{Bmass} < 3.85$ or $3.89 < \texttt{Bmass} < 3.92$ GeV

MC-constrained signal shape in the X baseline fit

PbPb24 signal MC



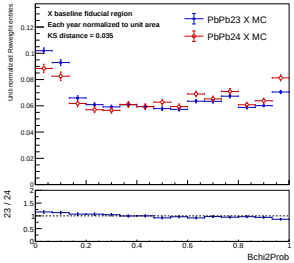
PbPb24 DATA



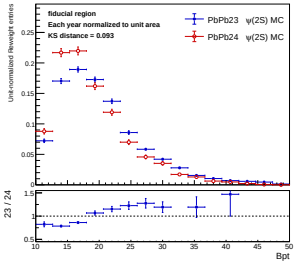
- Fit each year's reweighted signal MC with a common-mean double Gaussian.
- Simultaneous DATA fit: shared mean; year-specific yield, background and width scale.
- X baseline: MC widths and Gaussian fraction fixed; width scale in $[0.90, 1.50]$.

PbPb23 vs PbPb24 signal MC before ML

X: largest observed distance



$\psi(2S)$: largest observed distance



Channel	entries 2023 / 2024	mean / max distance	largest variable
X	56811 / 22257	0.0098 / 0.0346	`Bchi2Prob`
$\psi(2S)$	15717 / 7826	0.0389 / 0.0926	`Bpt`

X ML recipes tested (1/2)

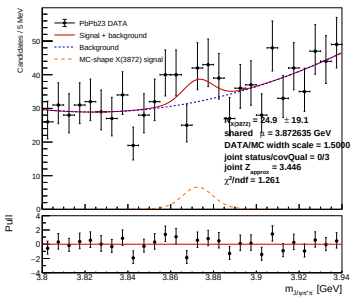
ML variables	best usable xeff	$\sqrt{q_0}$
Baseline (`Btrk1dR`, `Btktkpt`, `Btrk1Pt`, `Bchi2Prob`, `Bcos_dtheta`, `Btrk2Pt`)	25%	3.446
Baseline without `Btktkpt` (whole chain)	15%	3.512
Baseline without `Btktkpt` (ML-only)	10%	3.582
Baseline + `Bmu1y`	25%	3.770
Baseline + `Bmu2pt`	25%	2.849
Baseline + `Bmu1y` + `Bmu2pt`	30%	3.195
Baseline + `Bmu2y`	25%	3.922
Baseline + `Bmu1y` + `Bmu2y`	15%	3.853

X ML recipes tested (2/2)

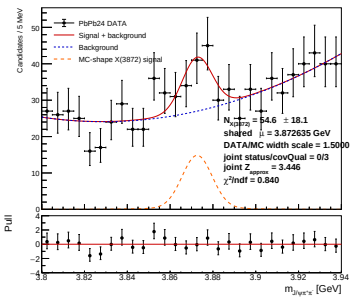
ML variables	best usable xeff	$\sqrt{q_0}$
Baseline + `Bmu1y` + `Bmu1pt`	20%	4.723
Baseline + `Bmu2y` + `Bmu2pt`	25%	3.506
Baseline + `Bmu1y` + `Bmu2y` + `Bmu1pt`	20%	4.407
Baseline + `Bmu1y` + `Bmu2y` + `Bmu1pt` + `Bmu2pt`	25%	4.616

- $q_0 = \max[0,\, 2\,(\mathrm{NLL}_{\mathrm{null}} - \mathrm{NLL}_{\mathrm{alt}})]$ from the simultaneous PbPb23+PbPb24 fit.
- Null fit: both signal yields fixed to zero; mean and width scales fixed at the alternative best fit; backgrounds reprofiled.
- The table reports $\sqrt{q_0}$ as a fit-only approximation at the best usable point in each seven-point scan.
- No toys, p_0 calibration or scan-trials correction is included; these values are not calibrated significances.

Baseline simultaneous fit at $x_{\text{eff}} = 25\%$



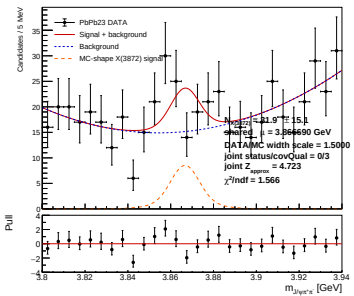
PbPb 2023 DATA



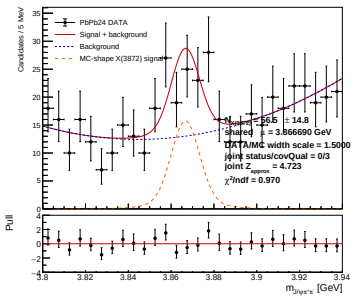
PbPb 2024 DATA

- Baseline: $\sqrt{q_0} = 3.446$; fitted yields 24.9 ± 19.1 and 54.6 ± 18.1.
- Both width scales reach 1.50.

Largest scanned proxy: baseline + Bmu1y + Bmu1pt



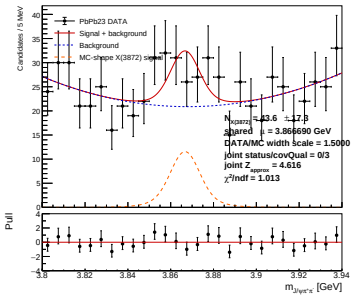
PbPb 2023 DATA



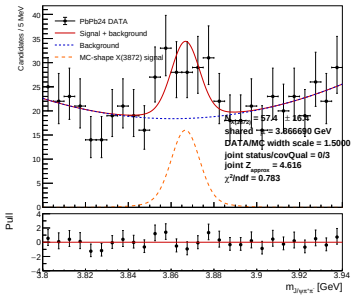
PbPb 2024 DATA

- At $x_{\text{eff}} = 20\%$: $\sqrt{q_0} = 4.723$; the largest value in the seven-point recipe scans.
- Shared mean and both width scales reach boundaries.

Baseline + four muon variables at $x_{\text{eff}} = 25\%$



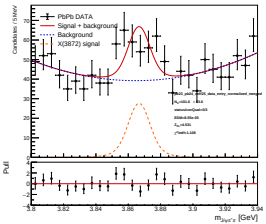
PbPb 2023 DATA



PbPb 2024 DATA

- Added variables: `Bmu1y`, `Bmu2y`, `Bmu1pt`, `Bmu2pt`; $\sqrt{q_0} = 4.616$.
- Shared mean and both width scales reach boundaries.

Direct merged fit as a stability estimate



Merged DATA fit
four-muon-variable recipe, $x_{\text{eff}} = 25\%$

- Normalize each year's weighted MC to its selected DATA entries:

$$w'_y = \text{Reweight} \times \frac{N_y^{\text{DATA}}}{\sum_i w_{y,i}}.$$

- Merge 2023 and 2024 MC and DATA; fit one signal yield, shape and background.
- The $N_{23}^{\text{DATA}} : N_{24}^{\text{DATA}}$ mixture is descriptive, not a physical signal-yield ratio.

Recipe and point	simultaneous	merged	2023 only	2024 only
+ `Bmu1y`, `Bmu1pt`; 20%	4.723	4.505	0.799	4.172
Baseline + four muon variables; 25%	4.616	4.531	2.633	3.792

Merged $\sqrt{q_0}$ is a same-point stability estimate, not a replacement for the nominal simultaneous fit.

Summary and Outlook

Topic	Current status
$\psi(2S)$ validation	PbPb23+PbPb24 workflow complete; combined signed $N_{\mathrm{eff}} = 54.2$ remains statistically limited.
Signal model	Year-specific reweighted-MC double-Gaussian shapes; shared DATA mean and independent yields/width scales.
Year comparison	Before ML selection, normalized X signal-MC shapes are similar between 2023 and 2024 (mean KS distance 0.010).
ML recipes	Muon-variable recipes give larger scanned local proxies; same-point merged fits provide a stability estimate.

Outlook:

- Try different reweighting and ML configurations to further optimize the X peak.

DRAFT – needs physics approval

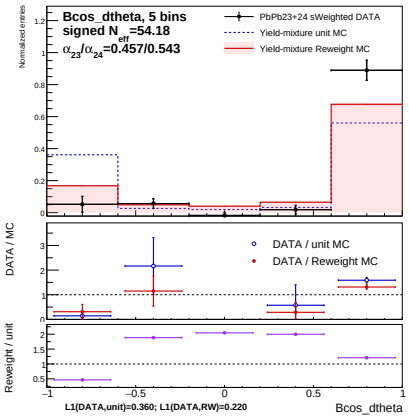
Thanks for listening!

Reporter: Lekai Yao

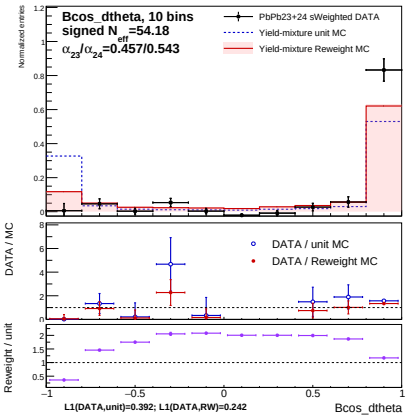
2026.8.27

$\psi(2S)$ validation: Bcos_dtheta

5 bins



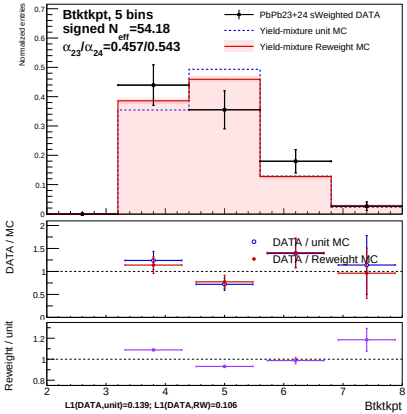
10 bins



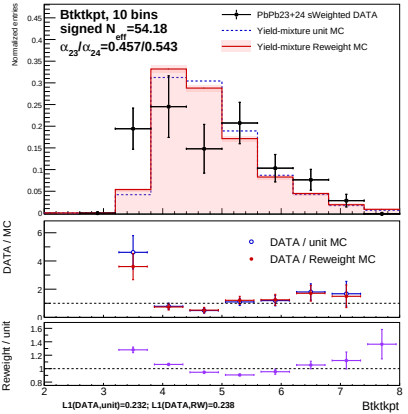
Reweighted-MC distance (L1/CDF): 5 bins 0.220/0.212; 10 bins 0.242/0.212.

$\psi(2S)$ validation: Btktkpt

5 bins



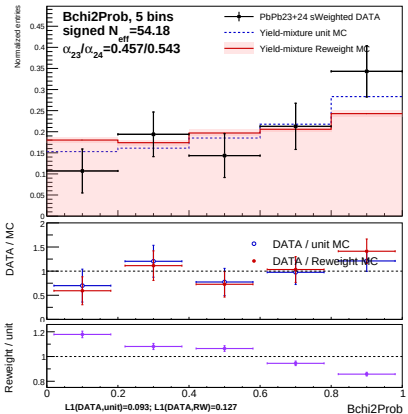
10 bins



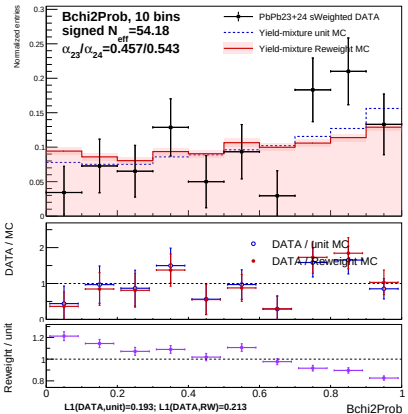
Reweighted-MC distance (L1/CDF): 5 bins 0.106/0.053; 10 bins 0.238/0.140.

$\psi(2S)$ validation: Bchi2Prob

5 bins



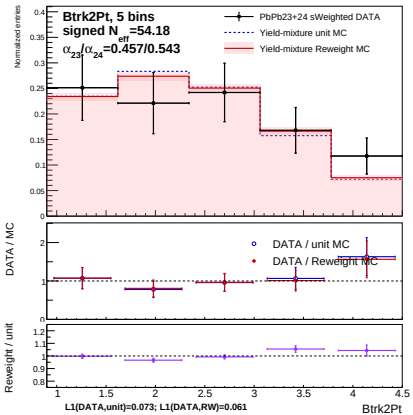
10 bins



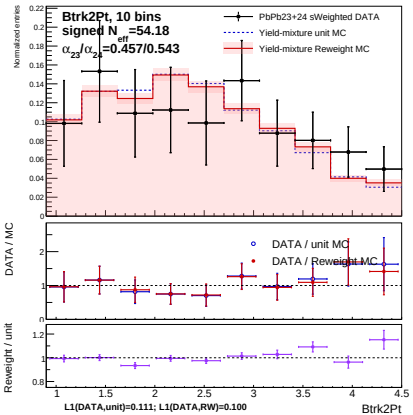
Reweighted-MC distance (L1/CDF): 5 bins 0.127/0.107; 10 bins 0.213/0.177.

$\psi(2S)$ validation: Btrk2Pt

5 bins



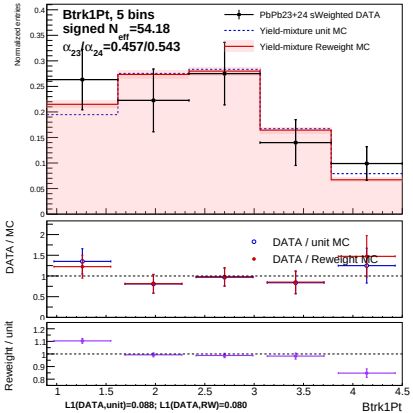
10 bins



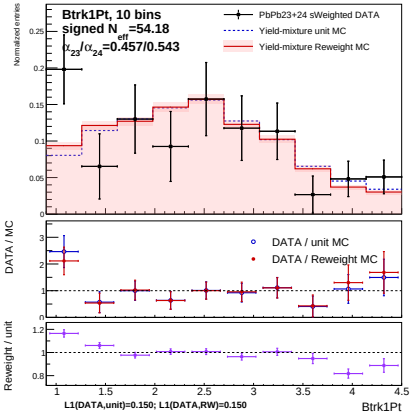
Reweighted-MC distance (L1/CDF): 5 bins 0.061/0.044; 10 bins 0.100/0.074.

$\psi(2S)$ validation: Btrk1Pt

5 bins



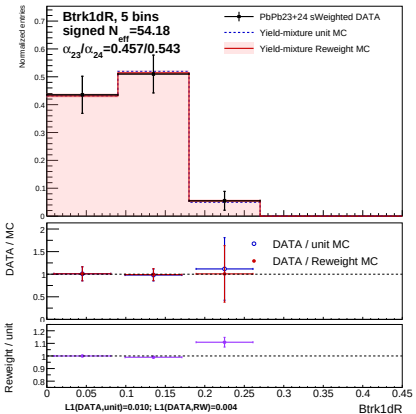
10 bins



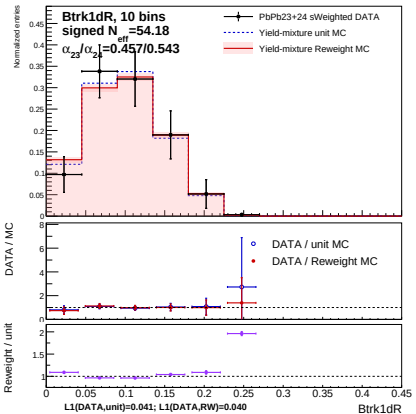
Reweighted-MC distance (L1/CDF): 5 bins 0.080/0.048; 10 bins 0.150/0.104.

$\psi(2S)$ validation: Btrk1dR

5 bins



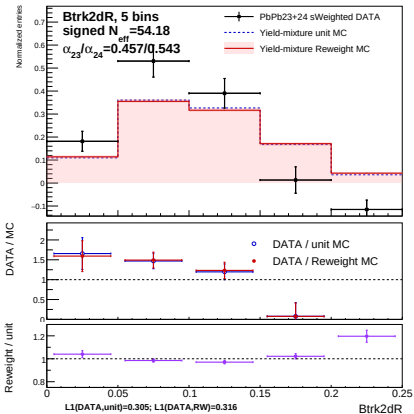
10 bins



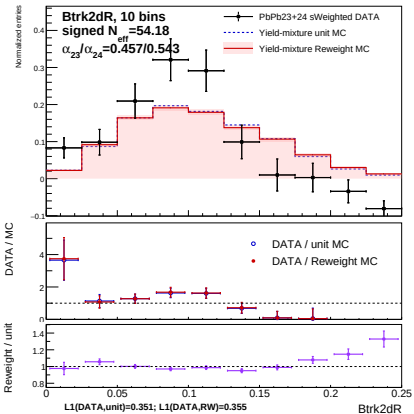
Reweighted-MC distance (L1/CDF): 5 bins 0.004/0.004; 10 bins 0.040/0.035.

$\psi(2S)$ validation: Btrk2dR

5 bins



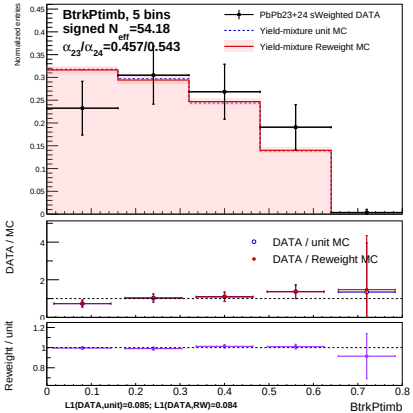
10 bins



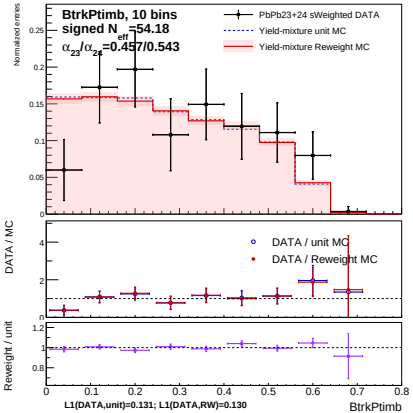
Reweighted-MC distance (L1/CDF): 5 bins 0.316/0.316; 10 bins 0.355/0.355.

$\psi(2S)$ validation: BtrkPtimb

5 bins



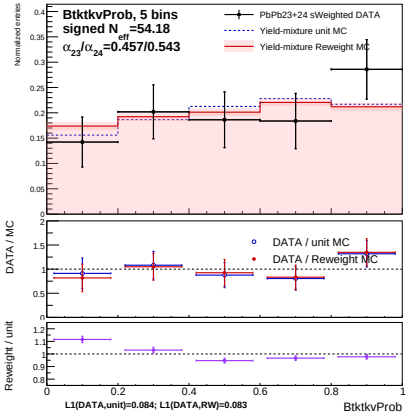
10 bins



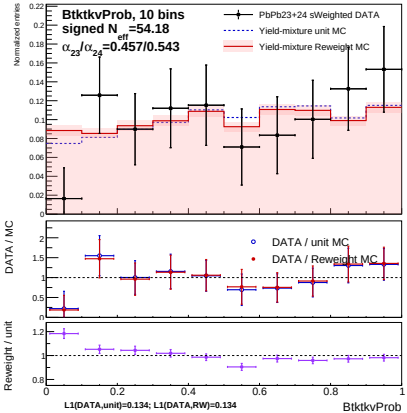
Reweighted-MC distance (L1/CDF): 5 bins 0.084/0.084; 10 bins 0.130/0.097.

$\psi(2S)$ validation: BtktkvProb

5 bins



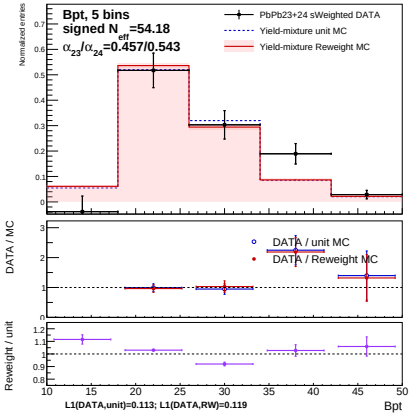
10 bins



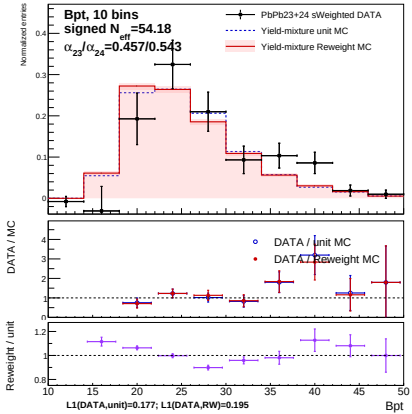
Reweighted-MC distance (L1/CDF): 5 bins 0.083/0.074; 10 bins 0.134/0.074.

$\psi(2S)$ validation: Bpt

5 bins



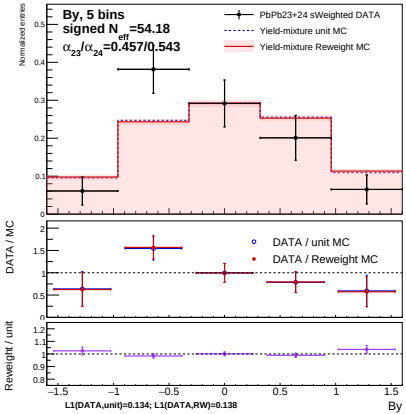
10 bins



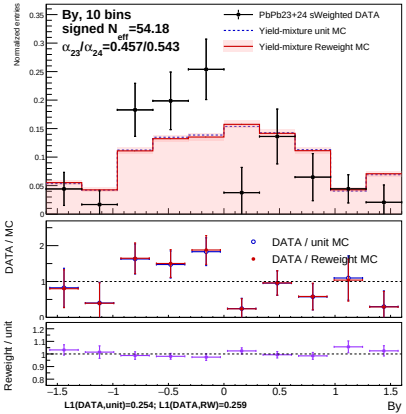
Reweighted-MC distance (L1/CDF): 5 bins 0.119/0.119; 10 bins 0.195/0.179.

$\psi(2S)$ validation: By

5 bins



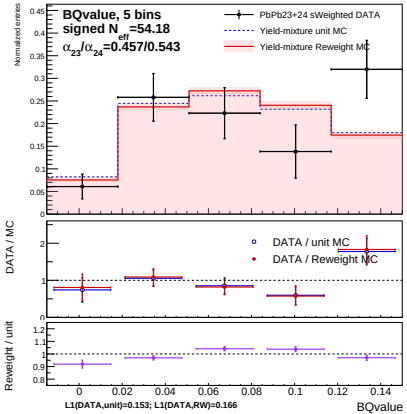
10 bins



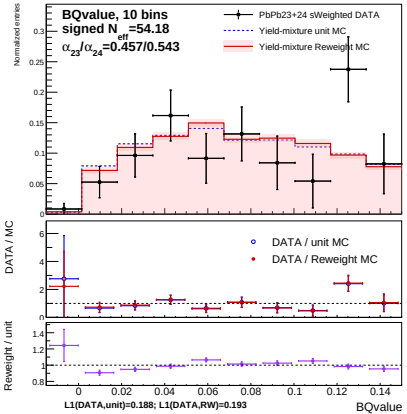
Reweighted-MC distance (L1/CDF): 5 bins 0.138/0.101; 10 bins 0.259/0.220.

$\psi(2S)$ validation: BQvalue

5 bins



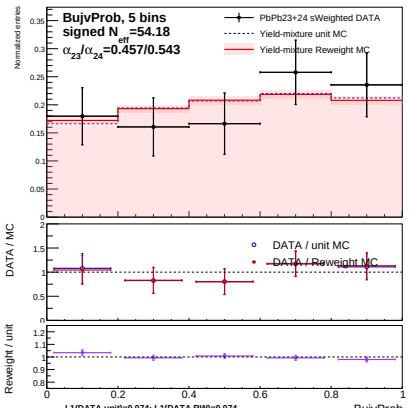
10 bins



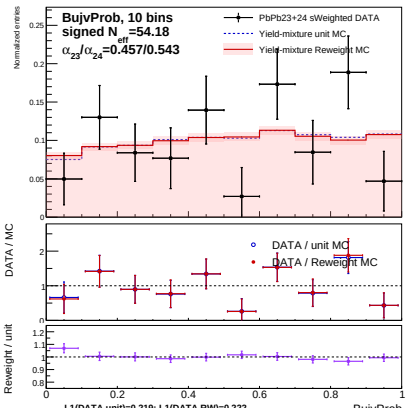
Reweighted-MC distance (L1/CDF): 5 bins 0.166/0.145; 10 bins 0.193/0.145.

$\psi(2S)$ validation: BujvProb

5 bins



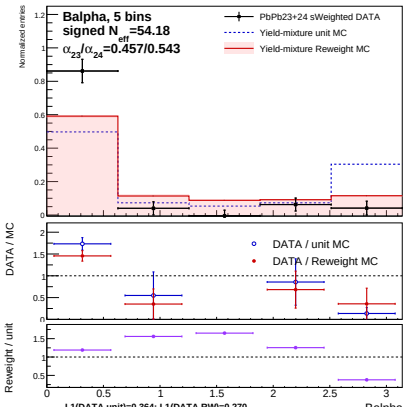
10 bins



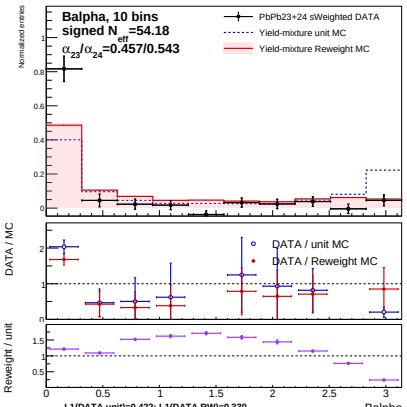
Reweighted-MC distance (L1/CDF): 5 bins 0.074/0.067; 10 bins 0.222/0.067.

$\psi(2S)$ validation: Balpha

5 bins



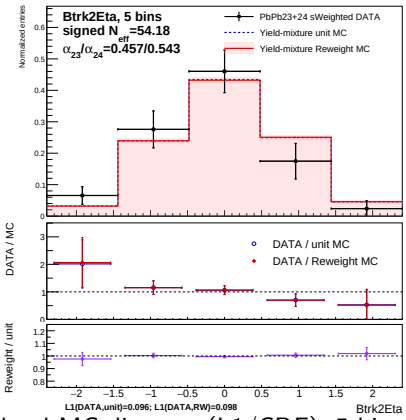
10 bins



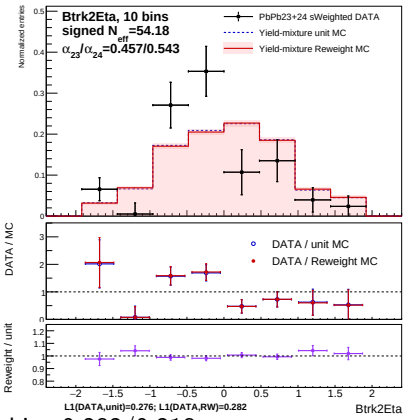
Reweighted-MC distance (L1/CDF): 5 bins 0.270/0.270; 10 bins 0.330/0.330.

$\psi(2S)$ validation: Btrk2Eta

5 bins



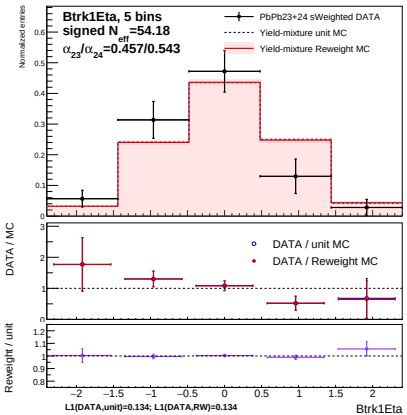
10 bins



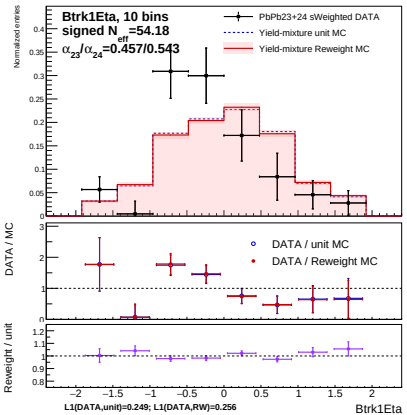
Reweighted-MC distance (L1/CDF): 5 bins 0.098/0.098; 10 bins 0.282/0.218.

$\psi(2S)$ validation: Btrk1Eta

5 bins



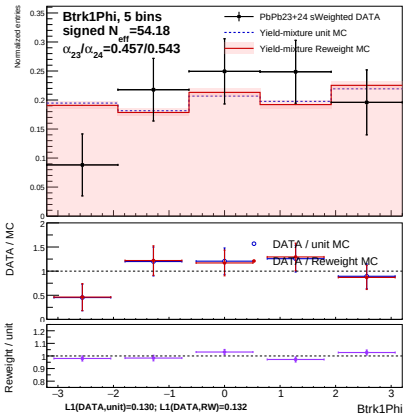
10 bins



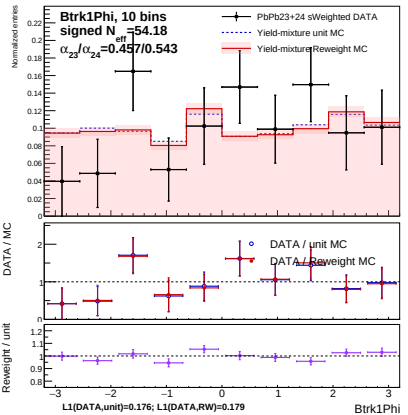
Reweighted-MC distance (L1/CDF): 5 bins 0.134/0.134; 10 bins 0.256/0.194.

$\psi(2S)$ validation: Btrk1Phi

5 bins



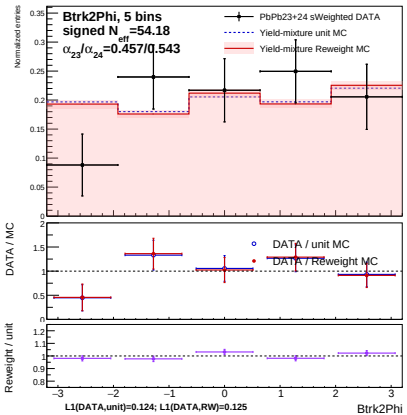
10 bins



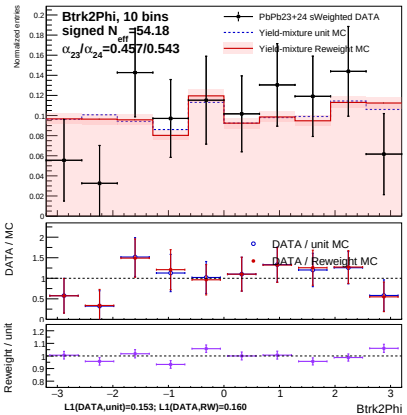
Reweighted-MC distance (L1/CDF): 5 bins 0.132/0.103; 10 bins 0.179/0.103.

$\psi(2S)$ validation: Btrk2Phi

5 bins



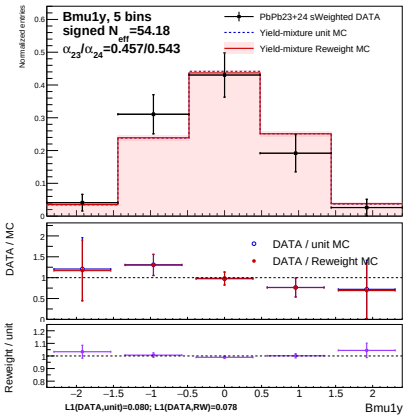
10 bins



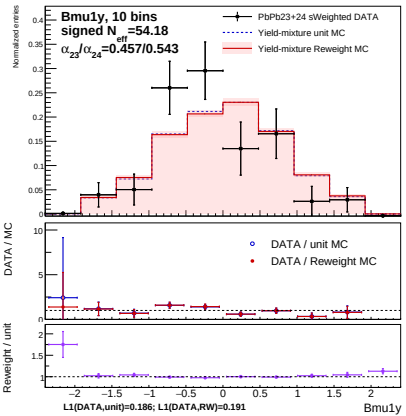
Reweighted-MC distance (L1/CDF): 5 bins 0.125/0.105; 10 bins 0.160/0.105.

$\psi(2S)$ validation: Bmu1y

5 bins



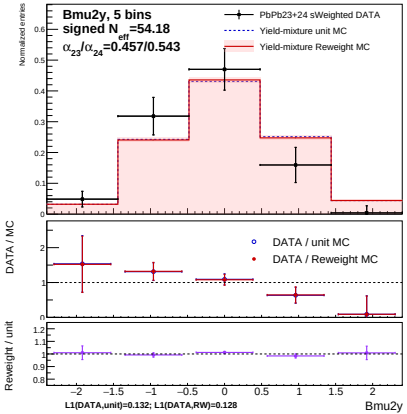
10 bins



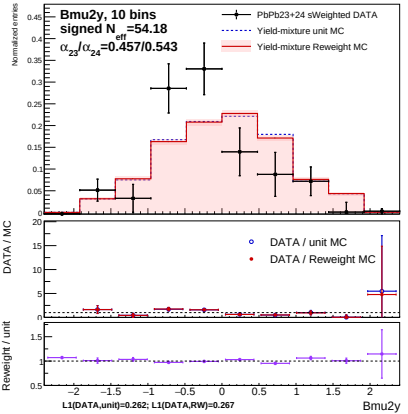
Reweighted-MC distance (L1/CDF): 5 bins 0.078/0.078; 10 bins 0.191/0.167.

$\psi(2S)$ validation: Bmu2y

5 bins



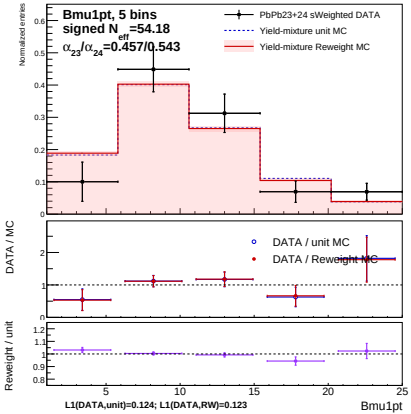
10 bins



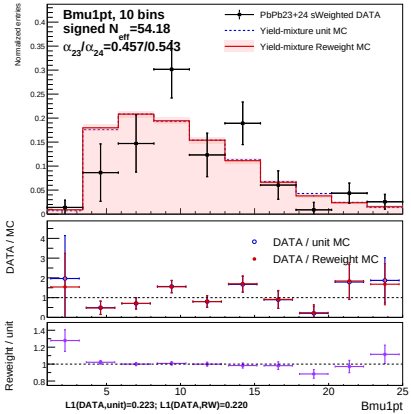
Reweighted-MC distance (L1/CDF): 5 bins 0.128/0.128; 10 bins 0.267/0.216.

$\psi(2S)$ validation: Bmu1pt

5 bins



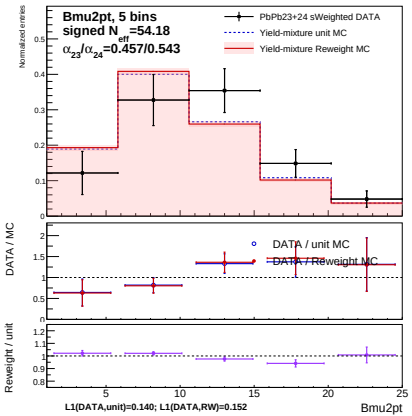
10 bins



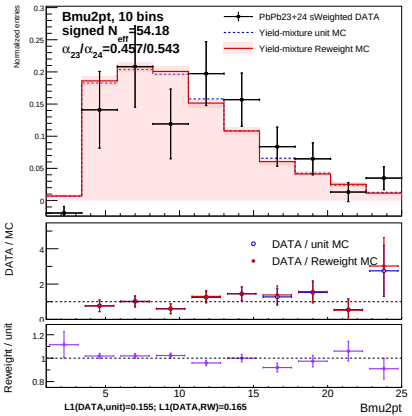
Reweighted-MC distance (L1/CDF): 5 bins 0.123/0.089; 10 bins 0.220/0.150.

$\psi(2S)$ validation: Bmu2pt

5 bins



10 bins



Reweighted-MC distance (L1/CDF): 5 bins 0.152/0.152; 10 bins 0.165/0.152.